

Reflection dynamic range

$\langle R_{\max} - R_{\min} \rangle$

0.6
0.5
0.4
0.3
0.2
0.1
0.0

Refractive index contrast $\Delta n = n_{\text{pillar}} - n_{\text{substrate}}$

$\Delta n_{\text{th}} \approx 0.5$

7× collapse
(Δn diff = 0.013)

*Flat spectrum
(no structural color)*

*Viable resonance
(structural color)*

Impedance
mismatch limit

- TiO₂
- a-Si
- ▲ Si₃N₄
- ◆ Al₂O₃

